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The Distribution of Three Saprotroph Orders Observed Against Costa Rica Hydrobasins

Question/Purpose:

The purpose of the three maps is to determine whether a correlation exists between three different orders (Xylariales, Auriculariales, and Helotiales) of macrofungi in Costa Rica and their distribution against the hydrological basins in this country using spatial analysis in ArcGIS Pro. A study on fungi from the order Aphyllophorales, was conducted in 2008 in order to determine a correlation between that and two different topographical spaces, a lakeside riparian forest and an upland boreal forest. After recording a total of 48 different species of wood fungi, researchers found that the fungi seemed to prefer the riparian forest as higher diversity and abundance levels were found in the lakeside riparian forest than the upland boreal forest (Komonen, 2008). This study found interesting results, but clarification from future research must be made in order to truly understand the ecological relationships at hand. For starters, elevation (i.e. sloped vs flat land) and land wetness levels were both taken into account in this study, so which variable actually has the correlation? The riparian zone was described as flat while the boreal forest was described as sloped (Komonen, 2008). Determining which variable(s) truly correlate to abundance and diversity levels in wood fungi are crucial in order to grow scientific understanding in the mycological field, and can be done so by testing each variable separately. My map's objective is to test the land wetness levels individually against fungal abundance.

A recent study, conducted in 2025, states Xylariales as a significant order of fungi to research due to its innate ability to produce secondary metabolite compounds with pharmaceutical and agrochemical uses. This study explores the biotechnological applications of these compounds from fungal species in the Xylariales order (Meshram, 2025). This study shows further significance and importance for studying the ecological behaviors of saprotrophic fungi like the order Xylariales. Uncovering more about this species gives depth to my map's objective by expanding it past the ecological realm. My map will hold purpose in the scientific community by displaying the relationship (if any) that exists between the locations of the hydrologic basins in Costa Rica and the distribution of the present fungi from the three orders, Xylariales, Auriculariales, and Helotiales. By doing so, one variable is singled out, making it easier to determine the existence of a relationship, and three very different orders of vital fungal species become further explored and understood. By the end, these maps should answer whether a correlational relationship exists between the distribution of Xylariales, Auriculariales, and Helotiales order of macrofungi and hydrological basins in Costa Rica?

Datasets:

My first dataset(s) displays the xy point coordinates of fungal distribution in Costa Rica. There are three of these datasets, one for each order (Auriculariales, Helotiales, Xylariales). They are CSV files that will be used as vector data in my map. The datasets are downloaded from filtered iNaturalist searches, and can only be referenced from the downloaded files themselves, however the links to obtain these csv files are cited in the References section below. I used these datasets to show the distribution of mushrooms by order in point form. Each of the three datasets includes data from a variety of years, therefore time/year is not a factor in my maps. These

datasets do not need any clipping because they were previously filtered to only include observations in Costa Rica.

The second main dataset displays the hydrological basins in Central America as polygons. There is only one of these datasets and it is used against each of the other three distribution datasets. Because the original dataset accounts for all of Central America, using the Clip tool is required to isolate the basins only in Costa Rica. This data was created in 2021, and therefore shows data from such time, but because the other three datasets do not include year, this will not be considered in the maps.

Results:

Five basins were identified for each order, and were chosen based on distribution counts within each basin. The five basins in each map with the highest counts of mushrooms observed were highlighted in each map. These numbers and rankings were determined by the tool, Summarize Within, which found the counts of mushrooms observed within each hydrobasin. Separate feature layers were made to showcase these high distribution hydrobasins called, dense_basins (or dense_hbasins, or dense_xbasins, or otherwise named after projections). For the Auriculariales order and map, we found the hydrobasins Tarcoles/Virilla/Colorado holding 23.8% of Auriculariales, Guacimal/Sardinal/Seco with 19.9%, Sarapiquí with 12.6%, San Carlos with 7.3%, and Parismina with 6.1% of the observed Auriculariales. These five basins had the highest amounts of observations of mushrooms within the Auriculariales order. For the Helotiales order and map, we found the hydrobasins Guacimal/Sardinal/Seco with 22.6% of Helotiales observed, Tarcoles/Virilla/Colorado with 18.9%, General with 17%, Parismina with 13.2%, and Parrita/Candelaria with 9.4% of observed Helotiales mushrooms. These five basins had the highest amounts of observations of mushrooms within the Helotiales order. For the

Xylariales order and map, we found the hydrobasins Parismina with 15.1% of observed Xylariales mushrooms, Tárcoles/Virilla/Colorado with 14.3%, General and San Carlos tied with 11.8%, and lastly the Guacimal/Sardinal/Seco with 8.4% of observed Xylariales. These five basins had the highest amounts of observations of mushrooms within the Xylariales.

As shown below, graphs were made for each order and map in order to further showcase the distributive findings of the datasets and Summarize Within tool. Each of these graphs show the results previously mentioned, but in the form of a pie chart. The usage of the pie chart graphing style was purposeful, as this mode best shows each of the five dense hydrobasins in each map/order.

Discussion:

The purpose of the maps were meant to answer whether a correlational relationship exists between the distribution of Xylariales, Auriculariales, and Helotiales order of macrofungi and hydrological basins in Costa Rica. Through the six maps and graphs created, we were able to find three hydrobasins in Costa Rica that contained some of the highest (within the top five) counts of observed Auriculariales, Helotiales, and Xylariales. The three basins found significant with all three orders include the Tarcoles/Virilla/Colorado, Guacimal/Sardinal/Seco, and the Parismina hydrobasins.

After researching through peer review studies to build my proposal for the map, I had found studies with inconclusive results due to multiple variable testing. Specifically, the distributions of mushrooms were tested against elevation and hydrobasins, but couldn't separate the data, which is why my maps and graphs only include distribution against hydrobasins. From my study, we can infer that saprotrophic mushrooms from the orders Auriculariales, Helotiales,

and Xylariales will most likely fruit within the Tarcoles/Virilla/Colorado, Guacimal/Sardinal/Seco, and the Parismina hydrobasins. However, further research is necessary to officially prove this determination.

In the future it would be beneficial to redo these maps with updated observation data, as well as with some extra maps and graphs. Three more maps and three more graphs, showing distribution against elevation, would be helpful to further determine which variable in the other studies had more significance or impact on the saprotrophic mushroom distribution. The updated observation data would also help legitimize the graphs and study with higher sample sizes.

References:

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